# PHYSICS - ELECTROSTATIC POTENTIAL AND CAPACITANCE

*Chapter 2 | Objective Questions | Class 12 | Maximum Time: 120 minutes*

## खण्ड अ — वस्तुनिष्ठ प्रश्न (Objective Type)

**1. If a uniform electric field exists along the X-axis, then the equi-potential surface will be**
(A) anywhere (B) in the YZ-plane (C) in the XY-plane (D) in the XZ-plane

**2. Work done in moving a charge between two points in an electrostatic field does not depend on the:**
(A) path taken (B) value of the charge (C) potential difference (D) shape of the conductor

**3. On inserting a dielectric material between the plates of a parallel plate capacitor, its capacitance**
(A) decreases (B) increases (C) remains unchanged (D) cannot be said

**4. For an isolated conductor, which of the following ratios is constant?**
(A) potential difference/total charge (B) None of these
(C) $\frac{\text{Total charge}}{\text{Potential}}$ (D) $\frac{(\text{Total charge})^2}{\text{Potential}}$

**5. The work done by the electric force in moving a charge along an equipotential surface is:**
(A) infinite (B) zero (C) maximum (D) negative

**6. Statement-I: The flow of negative charge is always from lower potential to higher potential. Statement-II: Electric potential is a vector quantity.**
(A) Statement-I is correct, but Statement-II is false.
(B) Both statements are correct but Statement-II is not the correct explanation of Statement-I.
(C) Both statements are correct and Statement-II is the correct explanation of Statement-I.
(D) Statement-I is false, but Statement-II is correct.

**7. An electrostatic field is—**
(A) and (B) Both (B) Conservative (C) None of these (D) Non-conservative

**8. Potential in the direction of the electric field:**
(A) first increases, then decreases (B) remains constant
(C) increases (D) decreases

**9. Plate X of a 2 μF capacitor is at 10 V and plate Y is at 20 V. The charge on plate X is—**
(A) $-10\mu C$ (B) $-20\mu C$ (C) Zero (D) $20\mu C$

**10. In a uniform electric field E, the magnitude of potential difference over distance d along the field is:**
(A) $E+d$ (B) $d/E$ (C) $Ed$ (D) $E/d$

**11. Statement-I: The capacitance of a capacitor increases with the increase of charge on its plate. Statement-II:** $V \propto Q$
(A) Statement-I is incorrect, but Statement-II is correct.
(B) Both Statement-I and Statement-II are correct, and Statement-II is the correct explanation of Statement-I.
(C) Both Statement-I and Statement-II are correct, but Statement-II is not the correct explanation of Statement-I.
(D) Statement-I is correct, but Statement-II is incorrect.

**12. Energy density of electric field**
(A) $3.6 \times 10^6$ J (B) Reduces alternating current
(C) Infinite (D) $\frac{1}{2}\epsilon_0 E^2$

**13. To find the total potential at a point due to two point charges, their potentials are:**
(A) added algebraically (B) multiplied
(C) compared only by magnitude (D) added as vectors

**14. Three capacitors, each of 4 μF, are connected in series between A and B. The equivalent capacitance is—**
(A) $2/3\mu F$ (B) $4\mu F$ (C) $3\mu F$ (D) $4/3\mu F$

**15. 64 identical drops, each of capacitance 5 μ F, combine to form a bigger drop. What will be the capacitance of the bigger drop?**
(A) 4 $\mu$F (B) 164 $\mu$F (C) 20 $\mu$F (D) 25 $\mu$F

**16. At a distant point, the potential due to an electric dipole is inversely proportional to:**
(A) $r^2$ (B) $\sqrt{r}$ (C) $r^3$ (D) $r$

**17. In the circuit, 3 μF and 2 μF capacitors are in parallel, and this combination is in series with a 4 μF capacitor. The equivalent capacitance between A and B is—**
(A) 1 $\mu$F (B) 20/9 $\mu$F (C) 9 $\mu$F (D) 1/9 $\mu$F

**18. Statement-I: The upper end of a lightning conductor installed on the roof of a building is pointed. Statement-II: This is done to increase the surface charge density.**
(A) Statement-I is incorrect, but Statement-II is correct.
(B) Statement-I is correct, but Statement-II is incorrect.
(C) Both statements are correct and Statement-II is the correct explanation of Statement-I.
(D) Both statements are correct but Statement-II is not the correct explanation of Statement-I.

**19. When dipole moment p is aligned parallel to a uniform electric field E, its potential energy is:**
(A) $-pE$ (B) $+pE$ (C) $p/E$ (D) zero

**20. 64 identical drops, each of capacitance $5\mu F$, combine to form a big drop. What will be the capacitance of the big drop?**
(A) $4\mu F$ (B) $25\mu F$ (C) $20\mu F$ (D) $164\mu F$

**21. The intensity of the electric field at any point on the surface of a charged conductor is:**
(A) Is perpendicular to the surface (B) Is zero
(C) Is at 45° to the surface (D) Is tangential to the surface

**22. The magnitude of torque on a dipole in a uniform electric field is:**
(A) $p/E$ (B) $pE\sin\theta$ (C) $E/p$ (D) $pE\cos\theta$

**23. In the figure, two 6 μF capacitors in series form one branch, in parallel with a single 6 μF capacitor. The equivalent capacitance between A and B is—**
(A) $1/9\mu F$ (B) $18\mu F$ (C) $6\mu F$ (D) $9\mu F$

**24. 64 identical drops, each of capacitance $5\mu F$, combine to form a bigger drop. What will be the capacitance of the bigger drop?**
(A) $164\mu F$ (B) $25\mu F$ (C) $20\mu F$ (D) $4\mu F$

**25. The total work around a closed path in an electrostatic field is:**
(A) positive (B) path-dependent (C) zero (D) negative

**26. Three capacitors, each of capacitance C, are connected in series. The resultant capacitance will be**
(A) 3C (B) 3/C (C) 1/3C (D) C/3

**27. Two capacitors $C_1 = 2\mu F$ and $C_2 = 4\mu F$ are connected in series and a potential difference (p.d.) of 1200V is applied across it. The Potential difference across $2\mu F$ will be:**
(A) 400V (B) 600V (C) 900V (D) 800V

**28. In an electrostatic field, potential difference between two points is defined as:**
(A) force per unit distance (B) work done per unit charge
(C) product of charge and distance (D) work done per unit time

**29. The electric potential due to a small electric dipole at a large distance r from the center of the dipole is proportional to:**
(A) $\frac{1}{r^3}$ (B) $r$ (C) $\frac{1}{r^2}$ (D) $\frac{1}{r}$

**30. Minimum number of capacitors of $2\mu F$ each required to obtain a capacitance of $5\mu F$ will be:**
(A) 4 (B) 5 (C) 3 (D) 6

**31. The electric field inside a charged conductor in electrostatic equilibrium is:**
(A) maximum normal to the surface (B) parallel to the surface
(C) infinite (D) zero

**32. Two point charges of $+10\mu C$ and $-10\mu C$ are placed at a distance 40 cm in air. Potential energy of the system will be -**
(A) -2.25 J (B) 2.35 J (C) -2.35 J (D) 2.25 J

**33. In electrostatic equilibrium, excess charge on a conductor resides:**
(A) uniformly in its volume (B) only on its inner surface
(C) on its surface (D) only at its centre

**34. The electric potential in equatorial position of an electric dipole is -**
(A) $\frac{1}{4\pi\epsilon_0}\frac{p}{r}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{p}{r^2}$
(C) Zero (D) $\frac{1}{4\pi\epsilon_0}\frac{p\cos\theta}{r^2}$

**35. The stored energy of a capacitor charged to 100 V, is 1 J. Capacitance of the capacitor is -**
(A) $2 \times 10^{-2} F$ (B) $2 \times 10^{-4} F$ (C) $2 \times 10^4 F$ (D) $2 \times 10^2 F$

**36. The magnitude of electric field just outside a conductor surface is:**
(A) $2\sigma/\epsilon_0$ (B) $\sigma\epsilon_0$ (C) $\sigma/\epsilon_0$ (D) $\epsilon_0/\sigma$

**37. Charge on a capacitor of capacity $1\mu F$ connected to a source of 1 volt will be**
(A) 1C (B) $1\mu C$ (C) Infinite (D) Zero

**38. A capacitor of capacitance $50\mu F$ is charged with 10V. The energy of the capacitor is?**
(A) $5 \times 10^{-2} J$ (B) $2.5 \times 10^{-3} J$ (C) $2.5 \times 10^{-4} J$ (D) $1.2 \times 10^{-8} J$

**39. The surface charge density near a pointed part of a conductor is generally:**
(A) smaller (B) zero (C) larger (D) unchanged

**40. Van de Graff generator is a machine which generates**
(A) High Current (B) High Voltage
(C) Both high current and high voltage (D) Only Low current

**41. 64 identical drops each of capacity $5\mu F$ combine to form a big drop. Capacity of the big drop will be**
(A) $4\mu F$ (B) $25\mu F$ (C) $164\mu F$ (D) $20\mu F$

**42. Electrostatic shielding is based on the fact that inside a closed conducting shell:**
(A) potential is infinite (B) charge cannot be nonzero
(C) field depends only on time (D) electric field is zero

**43. In the circuit, 3 μF and 2 μF capacitors are in parallel, and this combination is in series with a 4 μF capacitor. The equivalent capacitance between A and B is—**
(A) $1\mu F$ (B) $9\mu F$ (C) $\frac{20}{9}\mu F$ (D) $\frac{1}{9}\mu F$

**44. The value of electric potential at a distance r from a point charge q is**
(A) $\frac{q.r}{4\pi\epsilon_0}$ (B) $\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}$
(C) $\frac{1}{4\pi\epsilon_0}\frac{q^2}{r}$ (D) $\frac{1}{4\pi\epsilon_0}\frac{q}{r}$

**45. The capacitance of an isolated conductor depends on its:**
(A) potential (B) shape and surroundings (C) charge (D) change in charge-to-potential ratio

**46. The dielectric constant of water is**
(A) 1 (B) 80 (C) 60 (D) 42.5

**47. Electrical capacitance of earth of radius R is**
(A) $4\pi\epsilon_0 R$ (B) $\frac{4\pi\epsilon_0}{R}$ (C) $\frac{R}{4\pi\epsilon_0}$ (D) $4\pi\epsilon_0 . R^2$

**48. If the plate separation of a parallel-plate capacitor is doubled, with other conditions unchanged, its capacitance:**
(A) remains unchanged (B) becomes four times (C) doubles (D) halves

**49. Electric-potential is equal to**
(A) $wq$ (B) $\frac{q}{w}$ (C) $\frac{w}{q}$ (D) $\sqrt{wq}$

**50. If the plate area of a parallel-plate capacitor is doubled, its capacitance:**
(A) becomes four times (B) doubles (C) remains unchanged (D) halves

**51. One farad is equal to**
(A) 1 $CV^{-1}$ (B) 1 CV (C) 1 $CV^2$ (D) 1 $CV^{-2}$

**52. Potential energy of a charged conductor is**
(A) $CV^2$ (B) $\frac{1}{2}CV^2$ (C) $\frac{1}{4}CV^2$ (D) $\frac{1}{3}CV^2$

**53. In a series combination of capacitors, each capacitor has the same:**
(A) magnitude of charge (B) potential difference (C) energy (D) capacitance

**54. The value of $\epsilon_r$ in air is**
(A) 1 (B) zero (C) $9 \times 10^9$ (D) infinity

**55. Two capacitors, each of capacitance C, are connected in series. Their equivalent capacitance is—**
(A) 2C (B) $\frac{2}{C}$ (C) $\frac{C}{2}$ (D) C

**56. In a parallel combination of capacitors, each capacitor has the same:**
(A) potential difference (B) charge (C) energy (D) plate area

**57. Potential gradient is equal to**
(A) $dx.dV$ (B) $\frac{dV}{dx}$ (C) $\frac{dx}{dV}$ (D) None of these

**58. Which of the following is correct for the capacity of a parallel plate capacitor ?**
(A) $\frac{\epsilon_0 d}{A}$ (B) $\frac{\epsilon_0 A}{d}$ (C) $\frac{d}{\epsilon_0 A}$ (D) $\frac{A}{\epsilon_0 d}$

**59. Capacitors of 2 µF and 3 µF are connected in parallel. Their equivalent capacitance is:**
(A) 1.2 µF (B) 2.5 µF (C) 6 µF (D) 5 µF

**60. Dimensional formula of R.C. is**
(A) $M^0L^0T^0$ (B) $M^0L^0T^{-2}$ (C) $M^0L^0T$ (D) $M^0L^0T^{-1}$

**61. Capacity of spherical condenser is**
(A) $C = 4\pi\epsilon_0\left(\frac{rR}{r-R}\right)$ (B) $C = 4\pi\epsilon_0\left(\frac{rR^2}{r-R}\right)$ (C) $C = 4\pi\epsilon_0\left(\frac{rR}{r+R}\right)$ (D) $C = 4\pi\epsilon_0\left(\frac{r^2R}{r-R}\right)$

**62. Capacitors of 3 µF and 6 µF are connected in series. Their equivalent capacitance is:**
(A) 3 µF (B) 18 µF (C) 9 µF (D) 2 µF

**63. The capacity of any conductor is**
(A) $C = q \cdot v$ (B) $C = q/v$ (C) $C = v/q$ (D) $C = \frac{1}{2}q \cdot v$

**64. Energy of a charged conductor is**
(A) $E = C \cdot V$ (B) $E = \frac{1}{2}C \cdot V$ (C) $E = \frac{1}{2}C^2V$ (D) $E = \frac{1}{2}CV^2$

**65. A 4 µF capacitor connected to a 12 V battery carries charge:**
(A) 48 µC (B) 3 µC (C) 0.33 µC (D) 16 µC

**66. If n capacitors of equal capacity $C_1$ are connected in parallel, the equivalent capacity will be**
(A) $C = n^2C_1$ (B) $C = n/C_1$ (C) $C = nC_1$ (D) $C = C_1/n$

**67. Dielectric strength for air is**
(A) $4 \times 10^6$ $Vm^{-1}$ (B) $10^6$ $Vm^{-1}$ (C) $3 \times 10^6$ $Vm^{-1}$ (D) $5 \times 10^6$ $Vm^{-1}$

**68. A 5 µF capacitor carries charge 2 µC. The potential difference across it is:**
(A) 2.5 V (B) 0.4 V (C) 10 V (D) 25 V

**69. Van de Graaff generator is an electrostatic machine which produces**
(A) Only high voltage (B) Only high current (C) Low current and low voltage (D) High current and high voltage

**70. The net charge on a charged capacitor is**
(A) 1 C (B) infinite (C) $1\mu C$ (D) zero

**71. The correct expression for energy stored in a capacitor is:**
(A) $2CV^2$ (B) $Q^2/C$ (C) $QV$ (D) $Q^2/(2C)$

**72. Three capacitors of capacitance $6\mu F$ are available. The minimum and maximum capacitances obtained are**
(A) $2\mu F, 12\mu F$ (B) $6\mu F, 18\mu F$ (C) $2\mu F, 18\mu F$ (D) $3\mu F, 12\mu F$

**73. The energy density in a parallel-plate capacitor is:**
(A) $\frac{1}{2}\epsilon_0 E^2$ (B) $\epsilon_0 E^2$ (C) $2\epsilon_0 E$ (D) $\frac{1}{2}\epsilon_0/E^2$

**74. The value of electric potential at a distance r from a point charge is**
(A) inversely proportional to $r^2$ (B) proportional to r (C) proportional to $r^2$ (D) inversely proportional to r

**75. 64 identical drops each of capacity $5\mu F$ combine to form a big drop. What will be the capacity of the big drop?**
(A) $164\mu F$ (B) $20\mu F$ (C) $25\mu F$ (D) $4\mu F$

**76. When a dielectric completely fills a capacitor that remains connected to a battery, which quantity stays constant?**
(A) charge (B) stored energy (C) capacitance (D) potential difference

**77. If the distance between the two charges is increased, then the electrostatic potential energy of the charges**
(A) remains the same (B) increases (C) may increase or decreases (D) decrease

**78. In broad-side-on position, the electric potential due to electric dipole is**
(A) zero (B) infinite (C) $\frac{1}{4\pi\epsilon_0}\frac{p}{r^2}$ (D) $\frac{1}{4\pi\epsilon_0}\frac{p}{r}$

**79. When a dielectric is inserted into a capacitor disconnected from a battery, which quantity stays constant?**
(A) capacitance (B) energy density (C) potential difference (D) charge

**80. An electron is accelerated to 5 volt potential difference. The energy gained by the electron will be**
(A) 5 joule (B) 5 watt (C) 5 erg (D) 5 eV

**81. The relation between electric field (E) and electric potential (V) is**
(A) $E = -\frac{dr}{dV}$ (B) $E = \frac{dV}{dr}$ (C) $E = -\frac{dV}{dr}$ (D) $E = \frac{dr}{dV}$

**82. When a dielectric of constant K completely fills a capacitor, its capacitance:**
(A) becomes K times (B) becomes zero (C) is divided by K (D) remains unchanged

**83. The electrostatic energy of the system made by two electric dipoles kept at a distance 'r' is proportional to**
(A) $r^4$ (B) $r^3$ (C) $r^2$ (D) none of these

**84. Picofarad is the unit of**
(A) electric charge (B) electric flux (C) intensity of electric field (D) electric capacity

**85. If the capacitance of a battery-connected capacitor is increased, the charge drawn from the battery:**
(A) decreases (B) remains unchanged (C) increases (D) becomes zero

**86. Capacity of any condenser does not depend upon**
(A) size of plates (B) shape of plates (C) charges on plates (D) distance between plates

**87. The capacity of a spherical conductor is $1.0\mu F$. Its radius will be**
(A) 9 km (B) 1.11 cm (C) 10 metre (D) 1.11 metre

**88. The dependence of the far-field potential on the axial line of an electric dipole is:**
(A) $1/r^3$ (B) $r^2$ (C) $1/r^2$ (D) $1/r$

**89. The dielectric constant of a metal is**
(A) -1 (B) infinity (C) 1 (D) 0

**90. Outside a uniformly charged spherical conductor, the potential varies as if all charge were:**
(A) zero in the volume (B) concentrated at the centre (C) at equal distance on the surface (D) at infinity

**91. Which of the following is blocked by a capacitor ?**
(A) DC (B) Both AC and DC (C) Neither AC nor DC (D) AC

**92. The angle between equipotential surface and electric lines of force is**
(A) 45° (B) 180° (C) Zero (D) 90°

**93. Inside a charged spherical conductor of radius R, with zero potential at infinity, the potential is:**
(A) infinite at the centre (B) equal to the surface potential (C) proportional to radius (D) zero

**94. Potential at distance X out side a charged sphere of radius R charged with charge Q is**
(A) $\frac{kQ}{X}$ (B) $\frac{kQ}{X^2}$ (C) $kXQ$ (D) $\frac{kQ}{R}$

**95. An $\alpha$-particle is accelerated by a potential difference 200 V. Increase in its energy will be**
(A) 800 eV (B) 400 eV (C) 100 eV (D) Zero

**96. The magnitude of electric field at distance r from a point charge q is:**
(A) $kq^2/r$ (B) $kq/r$ (C) $kr^2/q$ (D) $kq/r^2$

**97. What is the energy stored in a capacitor of capacity C when its potential is increased up to V.**
(A) $\frac{1}{2}CV$ (B) $CV$ (C) $\frac{1}{2}CV^2$ (D) $\frac{1}{2CV}$

**98. N equal spherical drops of equal Potential V are Combined to form a big drop. Potential of new big drop will be**
(A) $V \times N$ (B) $V$ (C) $\frac{V}{N}$ (D) $V \times N^{\frac{2}{3}}$

**99. At the midpoint between two equal positive charges, the electric potential is:**
(A) positive (B) negative (C) always zero irrespective of the charges (D) zero

**100. Electric field in between the plates of a parallel plate condenser is $10^5$ V/m. If charge on plates of condenser is 1 $\mu$c then force on each plate of condenser is**
(A) None (B) 0.05 N (C) 0.5 N (D) 0.005 N